

semiconductor suppliers such as Marvell, Broadcom, and AMD would potentially use.

In semiconductors, we always have this aspect of building features that deliver value to the customer. For example, we have something called drive strength programmability on the outputs. It allows us to change the output drive strength to match the strength of the PCB trace more closely, resulting in lower noise and higher performance. Failure to match the PCB trace impedance to the output of the device can lead to reflections and noise on the trace, causing devices not to work correctly or reliably. So, we can program the drive strength. For instance, if someone wants to drive two loads with our device instead of one, we will increase the drive strength to accommodate two loads.

What are the key customer segments for your precision timing solutions? Which are the top three?

A segment where our value is very clear is the aerospace and defence market, because it is subject to very extreme environmental conditions and needs a timing device to survive that environment. Another market is communications and data centre equipment, where the value we provide solves significant problems.

The third place is automotive, where there are a lot of innovations happening. The amount of electronics going into a car is increasing dramatically, so that is another area where we have a lot of value. Other segments include the industrial segment with applications such as programmable factory automation, programmable controllers, factory automation, servers, motors, etc, and the mobile IoT and consumer market with applications such as smartwatches, consumer cameras, displays, etc.

Are the customer segments in India similar to your global markets, or very different?

Since these are the segments where the problems are more challenging, they are pretty similar. The Indian market has witnessed a tremendous pickup of electronics activity in the last few years due to government policies and probably other geopolitical considerations. Based on India's current market, two-wheelers can benefit from our timing devices, and that may be a new market for us to explore and figure out what kind of value we deliver to them.

Where would you place India as a market? How important is it for you right now?

We are seeing a lot more activity and designs originating in India. Based on our recent interactions with customers, we believe we can provide something of more value to electronics customers in India—an alternative to what they have been using. So, we must talk to customers, show them data, and let them decide.

Do you have partners in India?

We are already collaborating with companies such as Univa, Avnet, Arrow, and RFMW, which is a distributor. They provide front-end support to our customers with design resources and basic technical-level help. For highly technical questions, we connect the distributor FAEs (field application engineers) and customers' engineers with our support staff in Ukraine or Santa Clara.

Are you looking for new partners apart from Arrow?

Regarding partners, we have a programming partner in India. They program our blank devices and provide samples to customers for just-in-time needs. We have a

semiconductor companies partnership program to whom we provide engineering technical support to solve difficult timing problems on their reference validation platforms or customer evaluation boards, so their customers can benefit from our product value. We also have and seek independent design houses (IDH) whom we support. These white box designs then take the form of OEM products. So, that is the level of partnership that we have established. The evaluation process involves their assessing whether our offers benefit them. Simultaneously, we assess factors like their business, customer connections, and overall value, determining the value for us to provide our services.

How do you see your MEMS devices evolving to support future trends, and how do they compare with the technology trends with respect to your competitors?

The performance requirements of timing devices will only get more stringent as we witness the shift in precision timing from 4G to 5G. We saw timing improve threefold from 100 gigabit Ethernet (GbE) to 800 GbE. There's an additional variable: electronics itself is changing, with faster transmission and lower latency, making the performance of the timing device more critical.

We believe that MEMS technology can meet these evolving demands. Quartz technology has done a tremendous job of keeping up for the last 100 years, but at some point, you must assess the rate of innovation and determine which technology will perform better in the future. We believe that MEMS technology has an advantage here. Our innovation is not stopping; we will continue to innovate at the same rate. So, for the future of timing devices, we see MEMS as the technologically superior choice. **EFY**